

Personal Reflection on the Final Project

This project helped me get closer to technologies I had never used before, like joblib, and gave me a much deeper understanding of NLP in practice. Working through guided notebooks is very different from actually building your own pipeline from scratch. Here, I had to think through real problems with my data and with the NLP functions I was building, instead of just following steps that were already laid out for me. Building the project in a modular way, with each function developed and tested in its own notebook before being exported into the final agent, taught me a lot about how real NLP systems are structured. I also ran into practical issues I hadn't considered before, like keeping the TF-IDF and train/test transformations consistent, avoiding data leakage, and making sure new, unseen text could be processed using the exact same feature structure the model was trained on. Building the conversational layer was the part of this project that pushed me the most, since it was not about a single NLP technique, but about designing the logic that decides which technique to use and when. Splitting this into three small functions (`classify_intent`, `process_query`, `generate_response`) instead of one large function made it much easier to test and debug each part on its own before connecting everything together. I also ran into a few practical bugs while building this part, such as accidentally overwriting a function name and causing infinite recursion, and forgetting to apply a function to each row of a DataFrame instead of the whole column at once. Fixing these issues taught me to test small pieces in isolation before trusting them inside a larger pipeline. Overall, this final project helped me understand that a working NLP system is not just a collection of accurate models, but also the logic that connects them together in a way a real user can actually interact with.